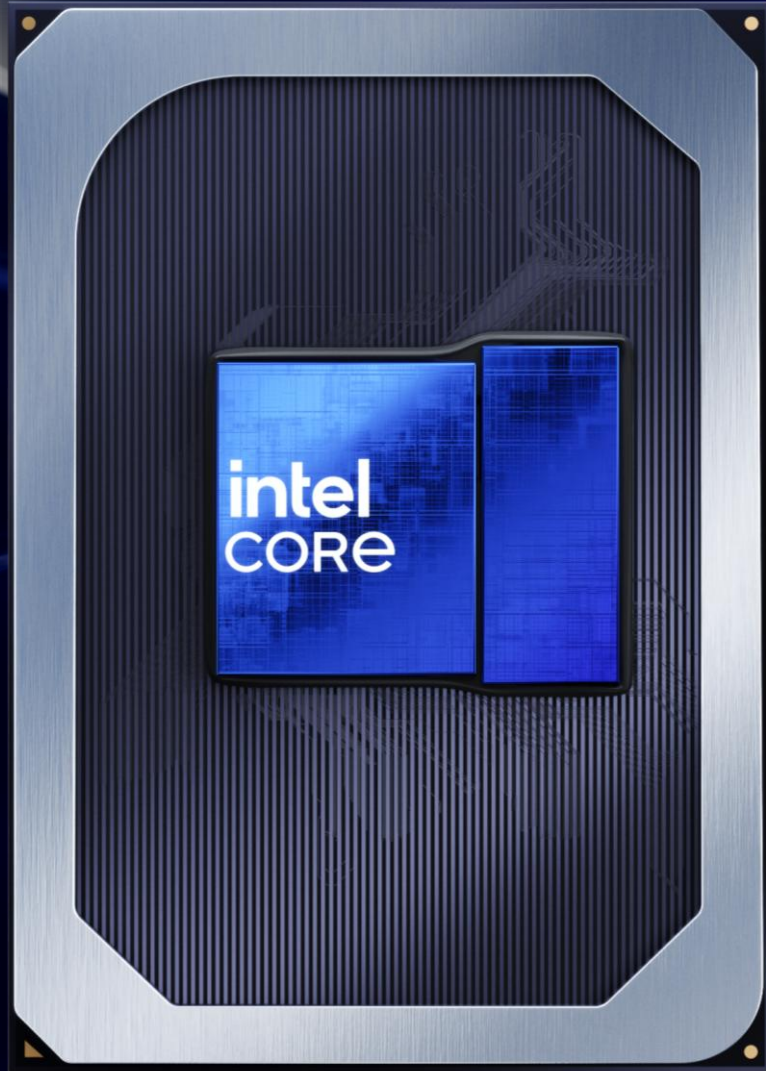




Hot Chips – August 2026

Intel® Core™ Series 3

Codename: Wildcat Lake (WCL)

Notices & Disclaimers

Performance varies by use, configuration and other factors. Learn more at intel.com/performanceindex.

Performance results are based on testing as of dates shown in configurations and may not reflect all publicly available updates. See backup for configuration details. No product or component can be absolutely secure.

Your costs and results may vary.

Intel technologies may require enabled hardware, software or service activation.

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Intel® Core™ Series 3

01

Built to maximize
value

02

SoC Construction
Trade-offs

03

Right Sized
Compute

04

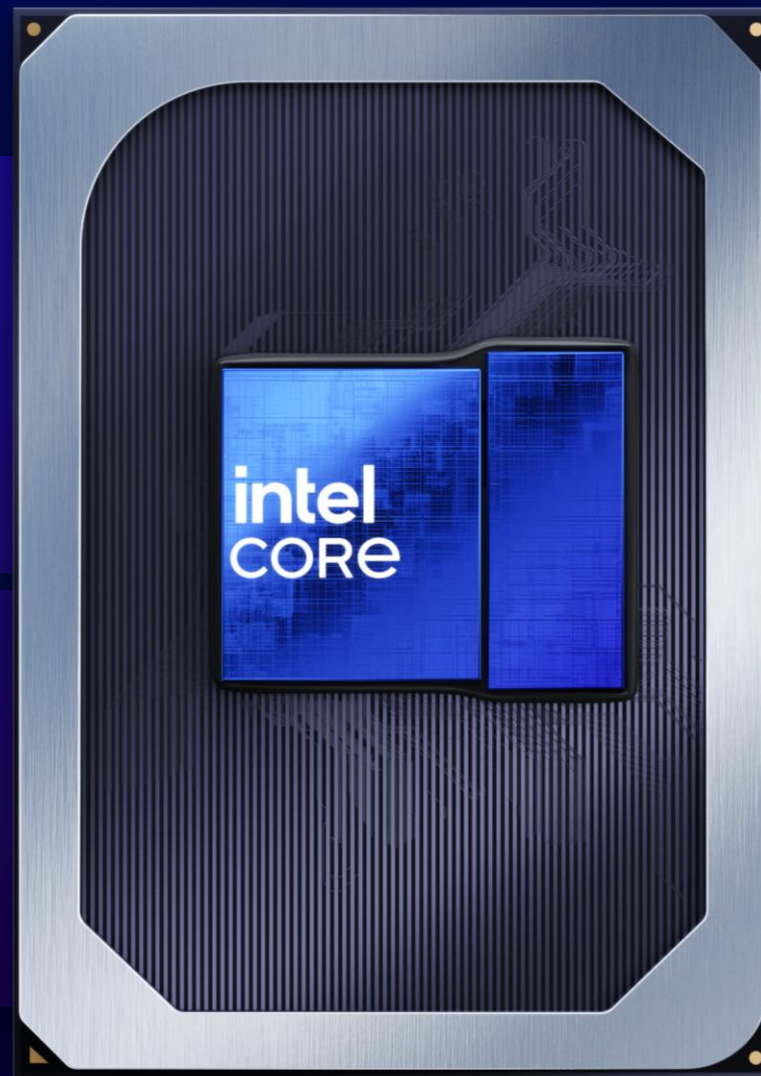
Right Sized
Platform

05

Key Innovation:
UCle for Client

06

Summary



Intel® Core™ Series 3

Overview



Derived from Core Ultra Series 3 (code name Panther Lake), designed to bring the features people care about most at broad scale



Modern connectivity without compromise, including Thunderbolt 4, Wi-Fi 7, and Bluetooth 6, so value systems still feel current and capable



Scales beyond notebooks to essential edge deployments, including robotics, smart buildings, point-of-sale terminals, and smart metering



AI-ready architecture delivering up to 40 platform TOPS for client AI workloads across CPU, GPU, and NPU

Up to **2.7X***
faster AI performance

Designed for all-day battery life in thin-and-light designs, improved efficiency for everyday productivity and creation

Up to **2.1X**
faster creation and productivity

Up to **64%**
lower processor power

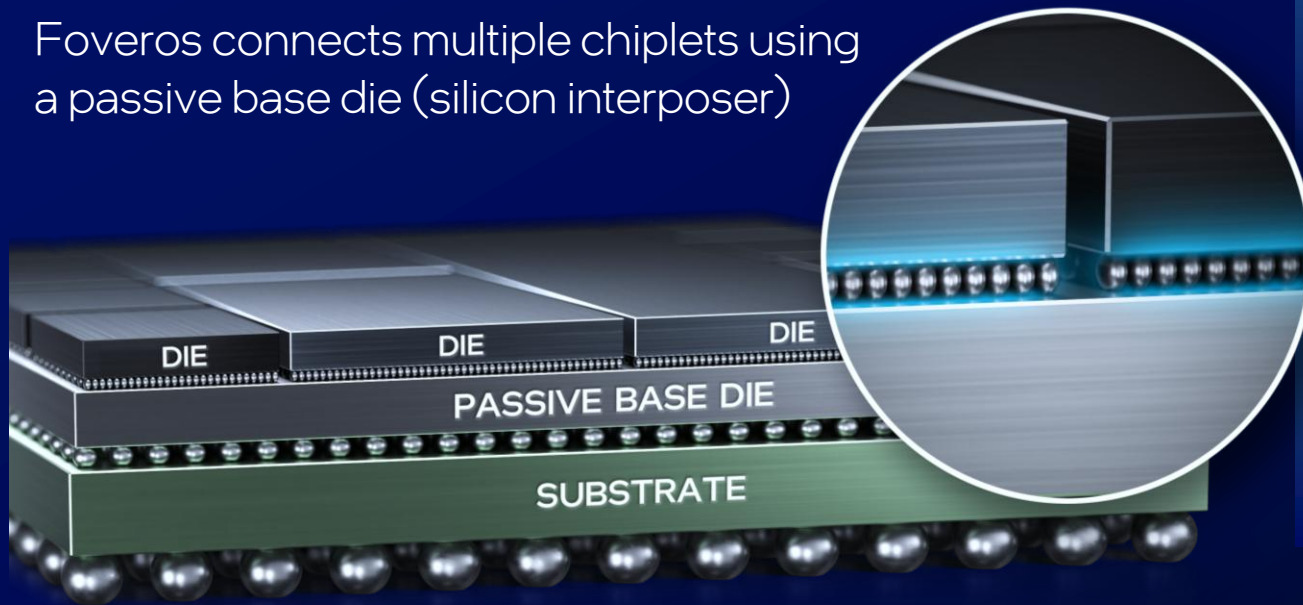
* Compared to Intel Core 7 Processor 150U (code name Raptor Lake U Refresh)

"Designed for All Day Battery Life": refers to laptops powered by Intel® Core™ Series 3 processors with minimum battery size and power efficient designs that leverage Intel's latest architecture, advanced compute technology and power optimizations that combine to deliver extended battery life while performing office multitasking, video playback, web browsing, and standby time in a typical consumer PC usage scenario and realistic environment. See [intel.com/performanceindex](https://www.intel.com/performanceindex) for workload and configuration details. Individual system results will vary significantly with different usages, battery capacity and other OEM design factors.

Packaging

Swap Foveros for MCP/Organic

Foveros connects multiple chiplets using a passive base die (silicon interposer)



Cost Benefits are Essential to this Segment

Elimination of the Base Die

Reduced Assembly cost and yield Loss

Requires Optimization to Maximize Benefits

Fewer Signals

Power Management

Must be **monolithic** or **MCP/Organic**

Path to Mainstream Success

From Intel® Core® Ultra Series 3

Primarily Drive cost reduction through

MCP

Right-size IPs

Right-size Platform

Maintain the Same Process Nodes & Die partition

Validation, schedule, & R&D

Best user experience
(ex: CPU ST perf, USB2 I/Os)

Highly focus innovation in a single area: UCle



Right Size Compute

for Mainstream

	Intel® Core® Ultra Series 3 “U”	Intel® Core® Series 3
Graphics	4Xe, 40 TOPs, Ray Tracing	2Xe Integrated, 20 TOPs
NPU	3 Tiles, 53 TOPs	1 Tile, 17 TOPs
P Cores	4, 12 MB L3	2, 6 MB L3
Memory	96GB, 128b L5-9.6, 8MB Cache	64GB, 64b L5-7.47, 4MB Cache
Camera	Yes	Removed
Media	8K60, w/ VVC decode & full effects	4K60, only HDR effect
Display	4x UHBR20	3x HBR3 (4K60)
		-38% Die Area

Right Size IO for Mainstream

	Intel® Core® Ultra Series 3	Intel® Core® Series 3
TBT/USB4	3 w/ eDP muxing (4 physical)	2
PCIe	4 Gen5, 8 Gen4	6 Gen4
USB	2x USB3.2, 8x USB2	2x USB3.2, 8x USB2
Camera PHYs	CSI	Removed
Audio	4 Speakers & Mics 5 DSPs 4.6 MB	2 Speakers & Mics 3 DSP (~0.8x due to higher frequency) 3.0 MB (0.8x)
Globals	Baseline	Few changes => few issues
		-15% Die Area

Cost Reduced Platform

Unique for Mainstream

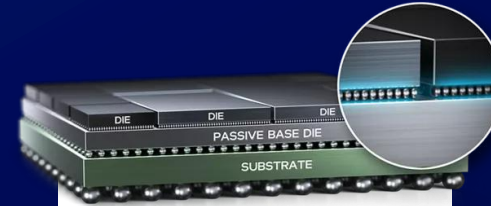
Enables Additional System Savings

Fewer IO for smaller package

Smaller Package saves board space.

Power Delivery Changes

No Dead ports



50x25x1.5 mm
Foveros

**Core Ultra
Series 3**

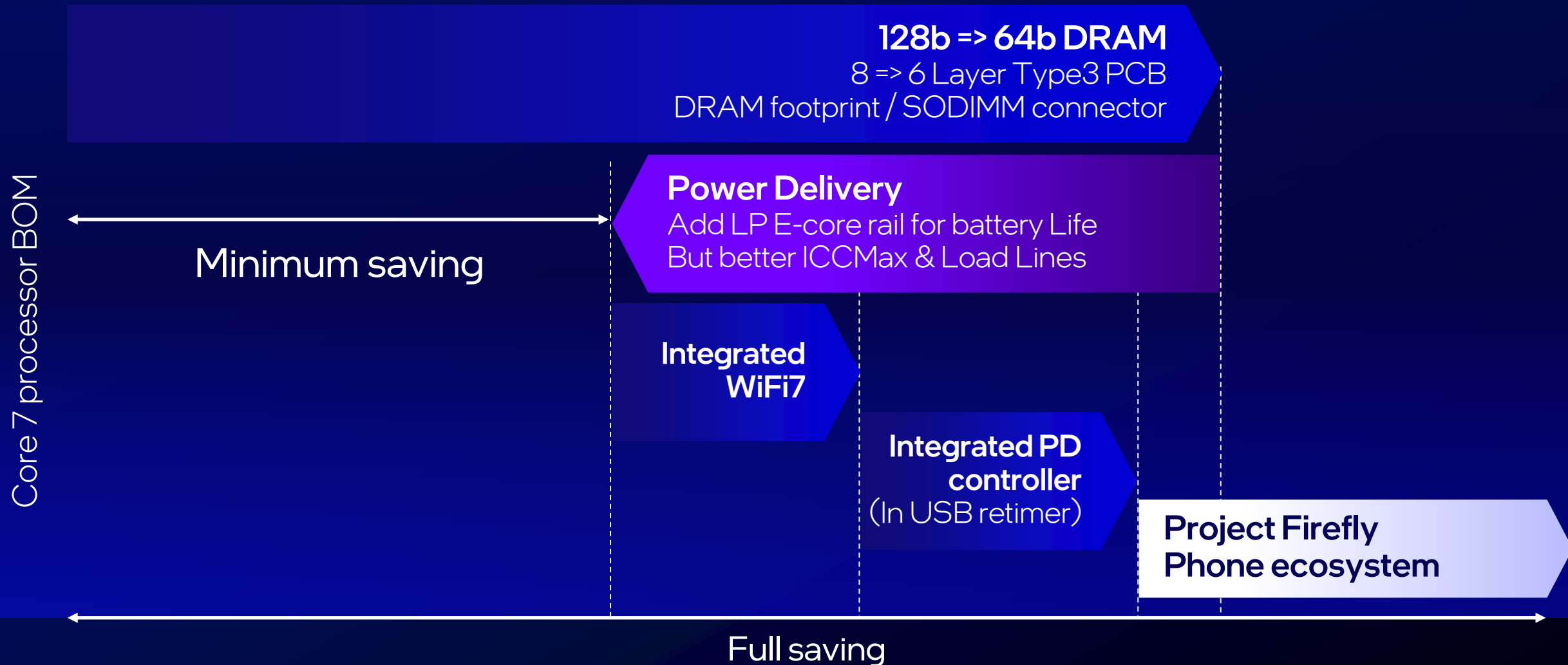
35x25x1.23 mm
Organic MCP

**Core
Series 3**



Platform Cost Savings

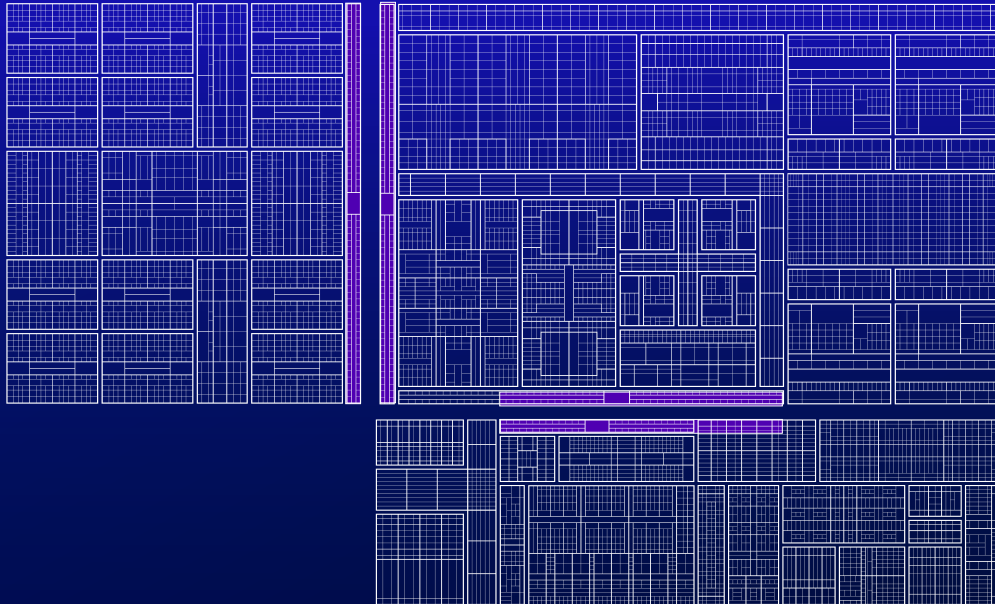
from Intel® Core® 7 processor 150U



MCP Savings Required a new Die to Die technology

UCle best fulfilled this need

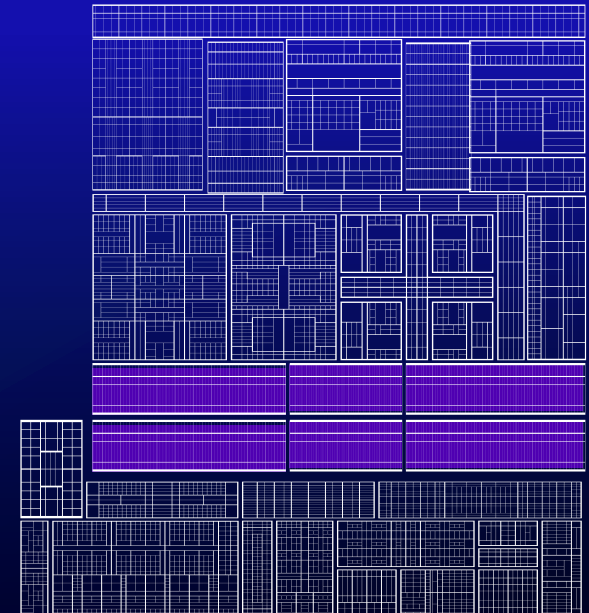
Intel® Core® Ultra Series 3 Die to Die



70% larger

Worth it for
MCP savings

Intel® Core® Series 3
Die to Die with UCle



UCle Area

No Base Die => Larger Bump Pitch

Foveros: 36 um

UCle: 110 um

Results

Need higher GTps

Bigger I/O area

Bigger Controller Area



Optimizations

Size IO bandwidth for Mainstream

Target 12 GBps for PCIe4 SSD
16R16W @ 8 GTps

Size Display: UHBR20 => 4Kp60

(Match RPL-U HDMI/eDP)
16W @ 8 GTps

Sideband Signals reduced
Many Packetized

UCle Protocol Changes

Data – Control Signal timing

Data now
packetized
(delayed)

Slower sideband
signals were not

UCle right in the middle of Boot, Debug,
Security, and Display protocols



Mitigations

Largest validation focus area

Fuses and survivability options
to ensure UCle bring up worked

UCle Power

Packetization increases Power

Clocks want to always run
to minimize latency hit

Idle systems still have Display traffic which
crosses UCle - especially without Panel Self
Refresh (PSR). Biggest power concern



Optimizations

Adopted multiple link states
Up to 8x idle power savings

QoS scheduling of Display traffic
Enables lower link states w/ display

UCle Signal Integrity

High Frequency Challenges

Tight voltage tolerances

Bit Error Rate (BER)



Optimizations

Initially had two IO Voltage options

- Board LDO: Cleaner, but more cost
- Use existing VCCAON rail
- Proved LDO not needed

Limited to 8 GTps max

- No Retry or Forward Error Correction
- Fall back options to lower frequency
- Less risk, but more area

Balanced Die Yield Recovery

Critical to cost, especially for a new process

	Pcores	LP Ecores	GPU	NPU
Config 1	2	4	2Xe	Yes
Config 2	1	4	1Xe	Yes
Embedded	2	4	1Xe	No

Frequency binning for CPU, GPU, and NPU

29% of compute die area recoverable

Any Pcore

Any Xe Core

NPU

Elected to maintain battery life and platform consistency and not recover these.

LP Ecore
(battery life)

Display Pipes
(consistency)

I/O
(consistency)

Intel® Core™ Series 3

A New Tier of Value for Mainstream

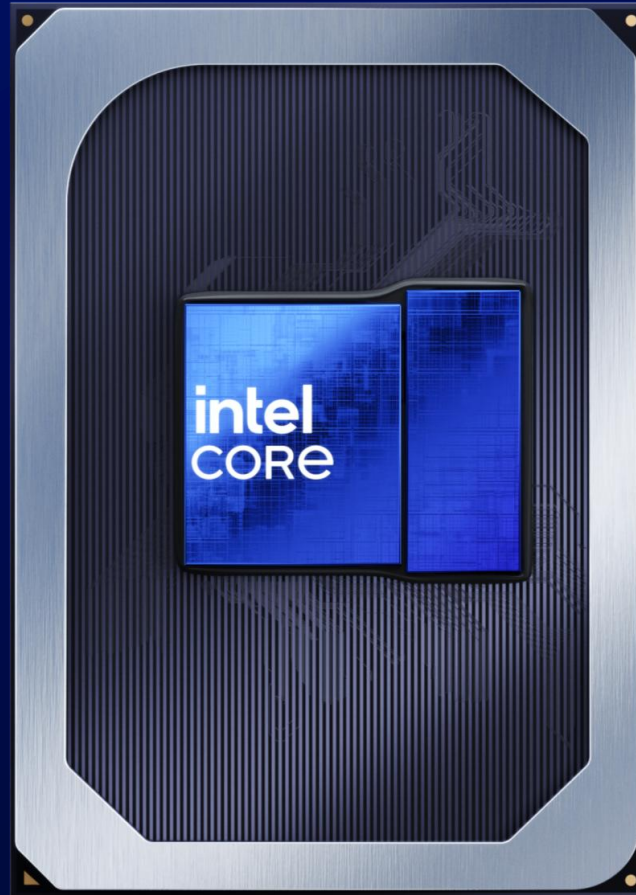
Cost Reduction

Foveros to MCP

Size IPs for Mainstream

Lower Platform Cost
(Package, BoM, IOs)

Reuse for Schedule & R&D



Excellent Value

Latest IP Technology
(not Gen N-2)

CPU Single Thread Perf

AI and DRAM Technology

Security and SoC Features

The Intel logo is centered on a dark blue background. It features a small, bright blue square positioned above the first vertical stroke of the lowercase letter 'i'. The word 'intel' is rendered in a clean, white, sans-serif typeface. A registered trademark symbol (®) is located at the end of the word, to the right of the final period.

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